

Apply a microtextured or hydrophilic antiadhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Post-operative endophthalmitis following intraocular lens implantation

Mechanism: Mechanical athalamia due to eye rubbing triggers forward IOL displacement, causing adhesion to the anterior capsule and trapping fluid.

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Apply a microtextured or hydrophilic antiadhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

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{'equation_id': 'FLUID-001', 'name': 'Hagen-Poiseuille (laminar pipe flow)', 'expression': 'Q = (pi * r^4 * dP) / (8 * mu * L)', 'source': {'text': 'standard Newtonian laminar pipe-flow relation (any fluid-mechanics text)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'fluidics_hydraulic'}, 'assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': []}, 'assumption_check': {'assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow', 'model_assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'FLUID-002', 'name': 'Reynolds number', 'expression': 'Re = (rho * v * D) / mu', 'source': {'text': 'standard dimensionless similarity number (any fluid-mechanics text)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'circular section; single-phase flow', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'fluidics_hydraulic'}, 'assumptions': ['transition criterion Re~2300 is a MODEL_DERIVED engineering rule of thumb'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': []}, 'assumption_check': {'assumptions': ['transition criterion Re~2300 is a MODEL_DERIVED engineering rule of thumb'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}}
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'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables':
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{'equation_id': 'FLUID-003', 'name': 'Orifice flow (sharp-edged restriction)', 'expression': ' $Q = C_d * A * \sqrt{2 * dP / \rho}$ ', 'source': {'text': 'standard orifice-equation form (any fluid-mechanics text)',
'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition':
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measurement'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL',
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until measured', 'model_assumptions': [' $C_d \sim 0.6-0.65$  is an engineering prior, not a measurement'],
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[], 'mechanism_mention_count': 0}}

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Kill Condition: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.